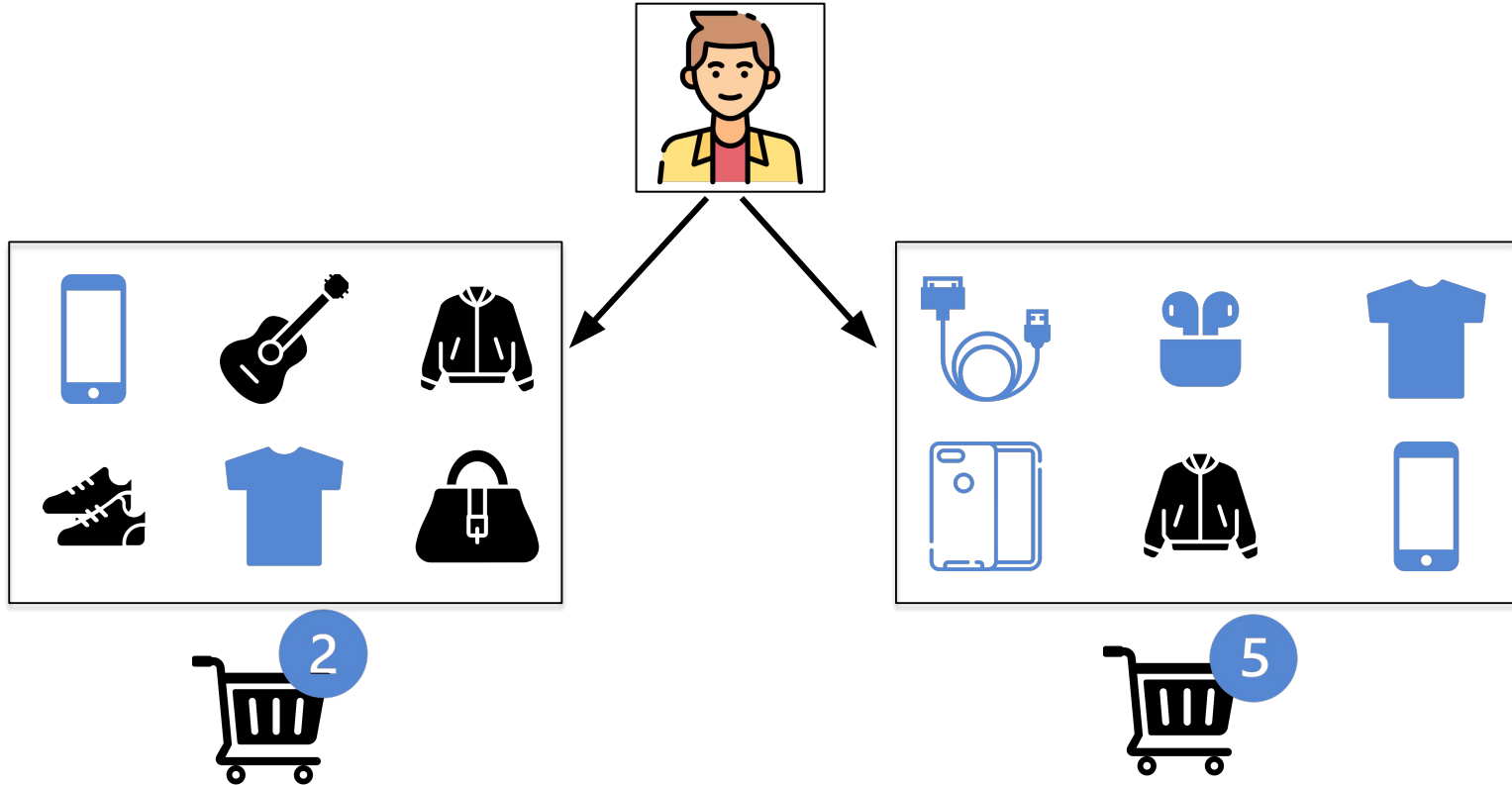


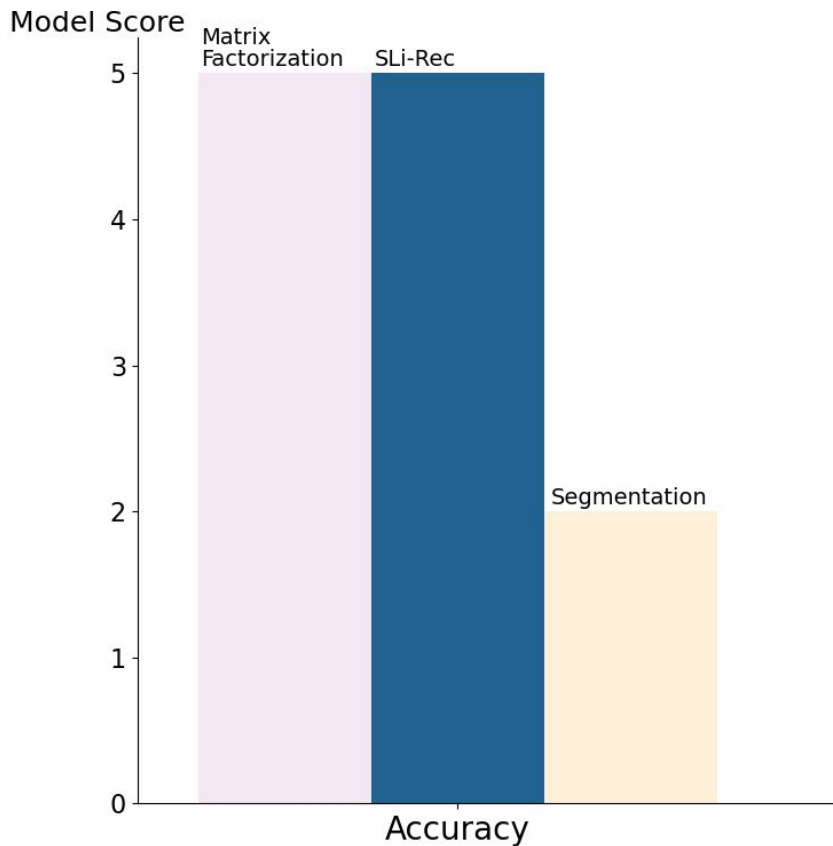
Enhancing Online Sales with Smarter Product Discovery

Leonard Eshun, Mobasserul Haque, Meron Gedrago, Diego Rodriguez

As our company explore new ways to **grow online sales**, one opportunity is to **facilitate product discovery**

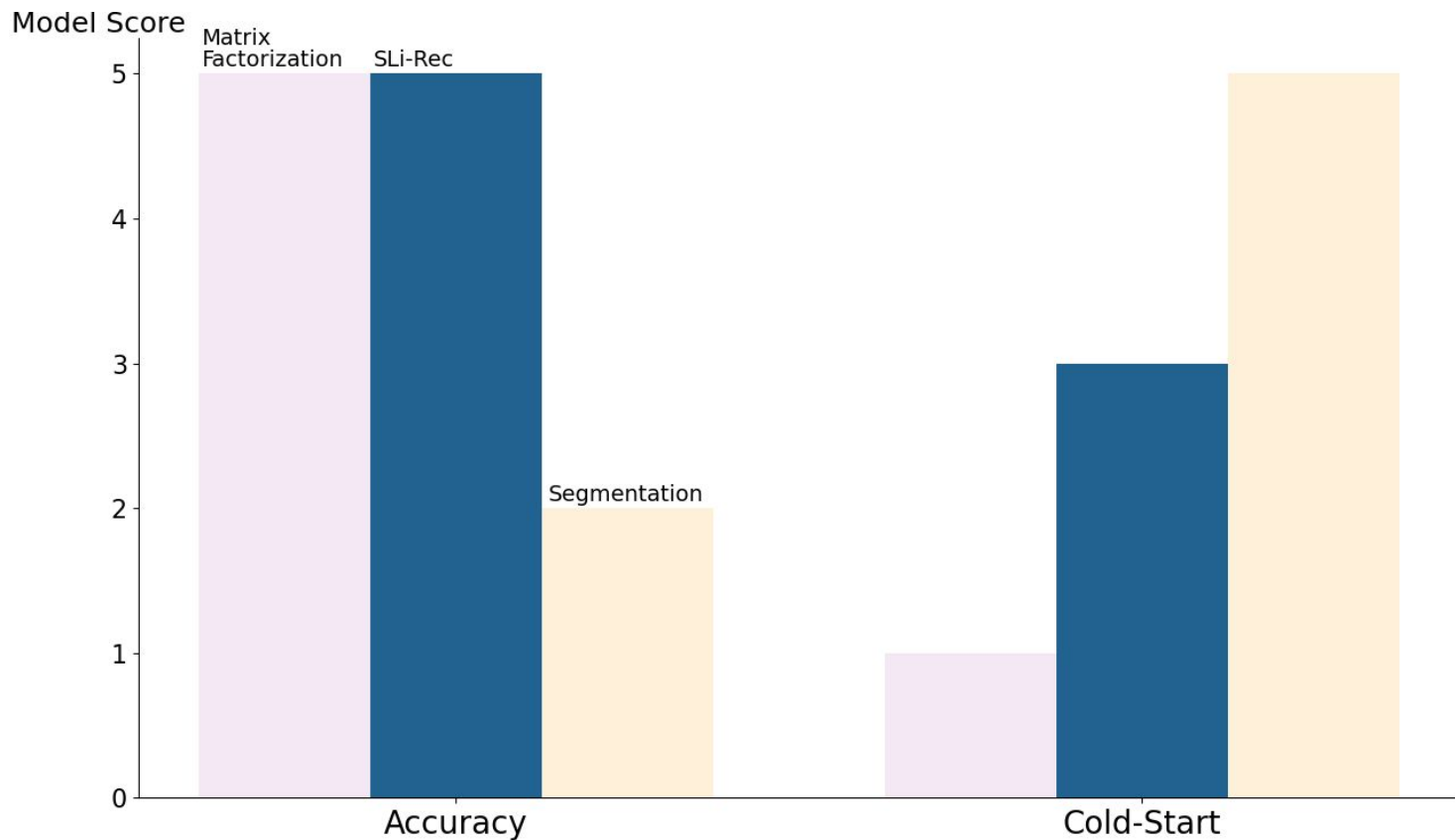


SLi-Rec has a balanced metrics trade-off



We chose SLi-Rec because it has a balanced trade-off for key metrics compared with other recommender techniques.

SLi-Rec has a balanced metrics trade-off



We chose SLi-Rec because it has a balanced trade-off for key metrics compared with other recommender techniques.

The **Blind Spot**: Missing Context Leads to **Repetitive Recommendations**



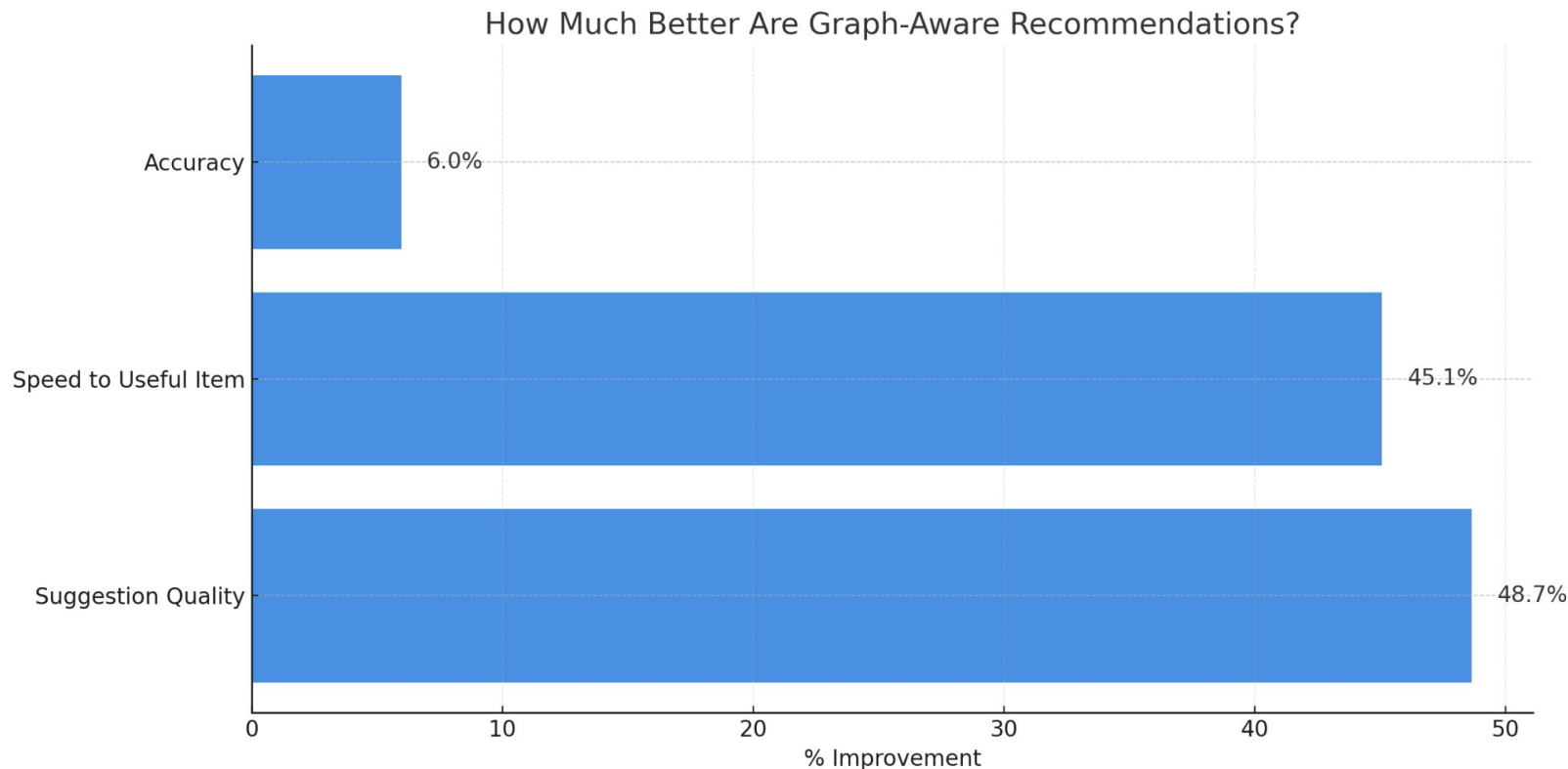
Without product relationships, the model can only repeat what it sees

Adding Product Relationships Gives the Model the Missing Context

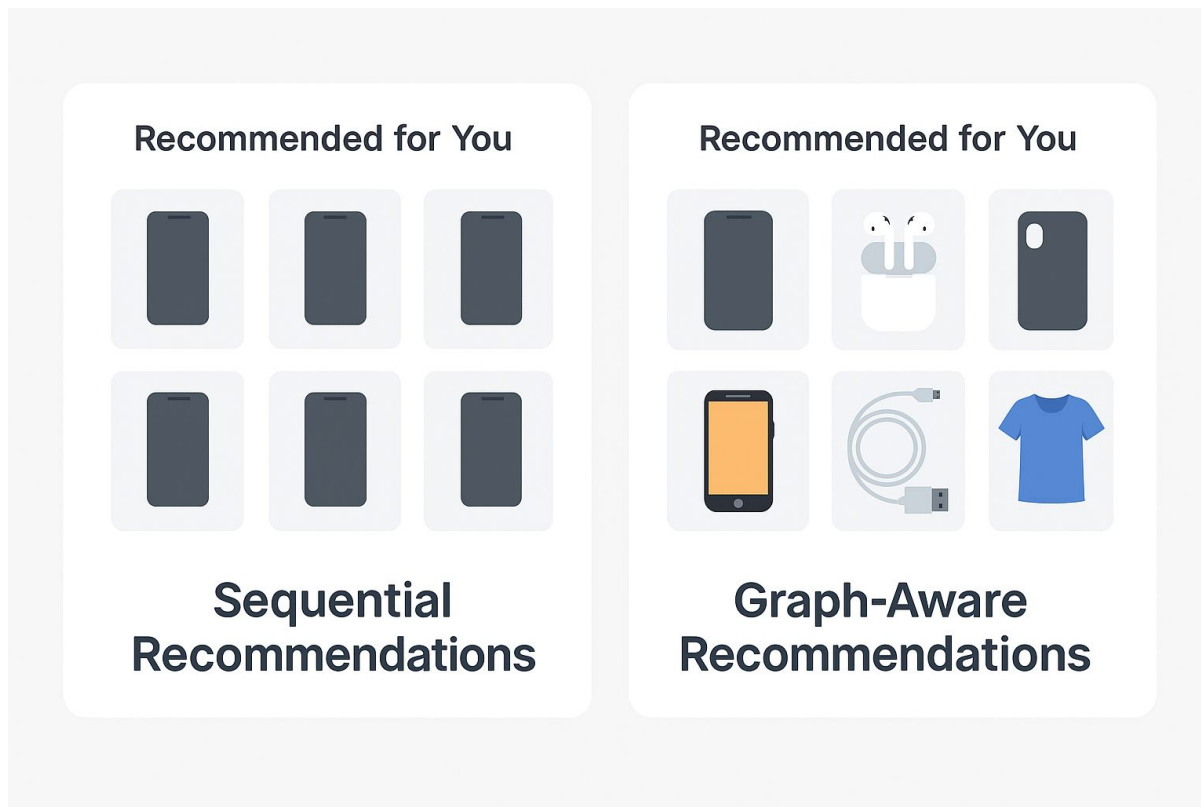


We can push engagement further by adding more context into the model

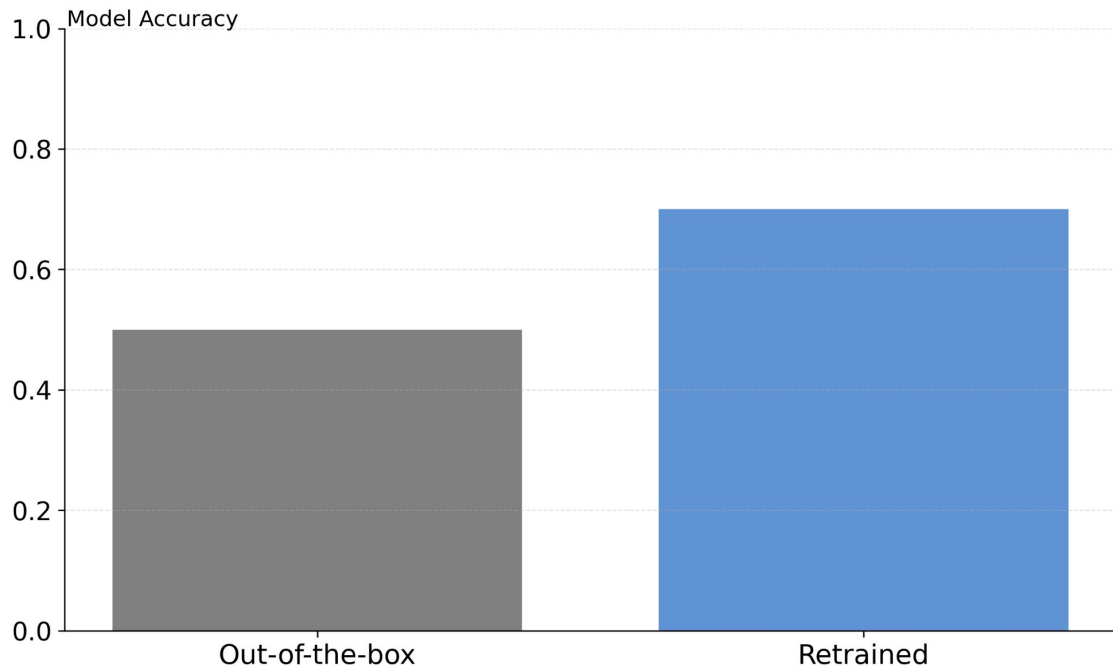
Context boosts accuracy, helps users find useful items faster, improves overall suggestion quality.



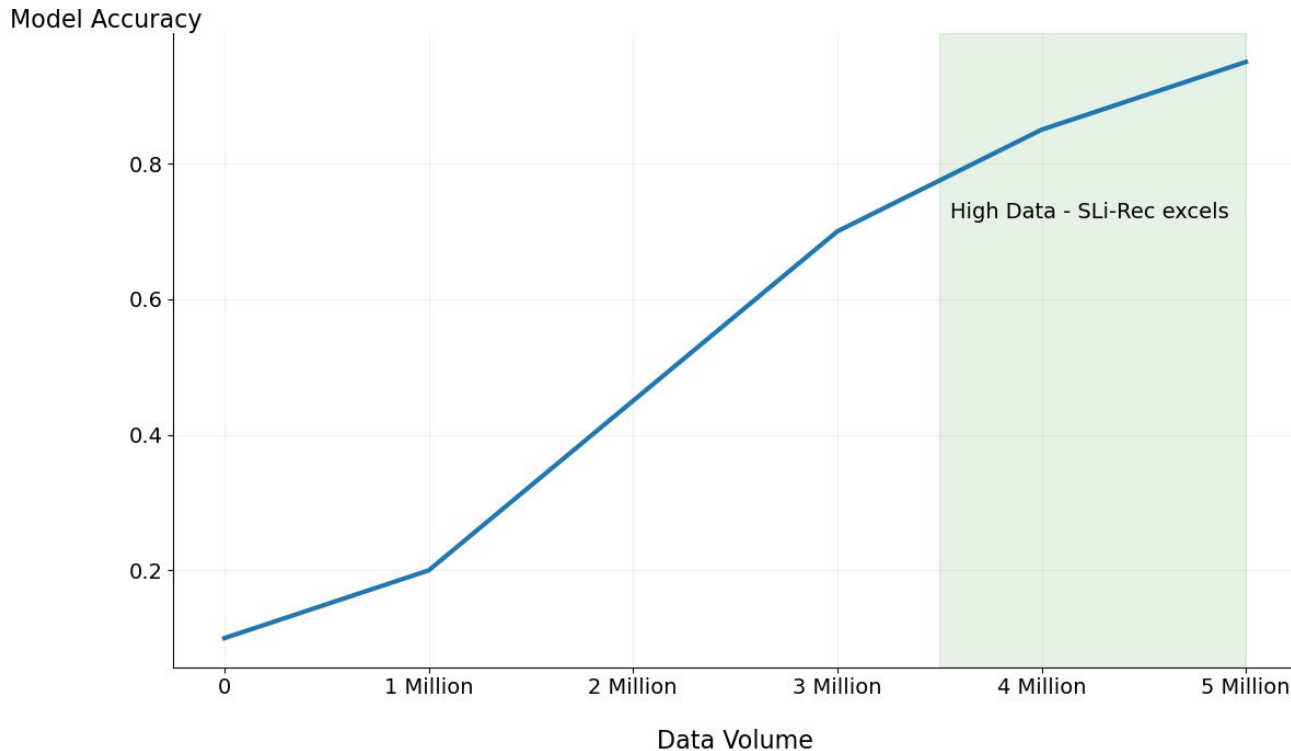
The Result: **Less Repetition**, **More Relevant Options**



Re-training models on specific product-lines can improve model performance and recommendations

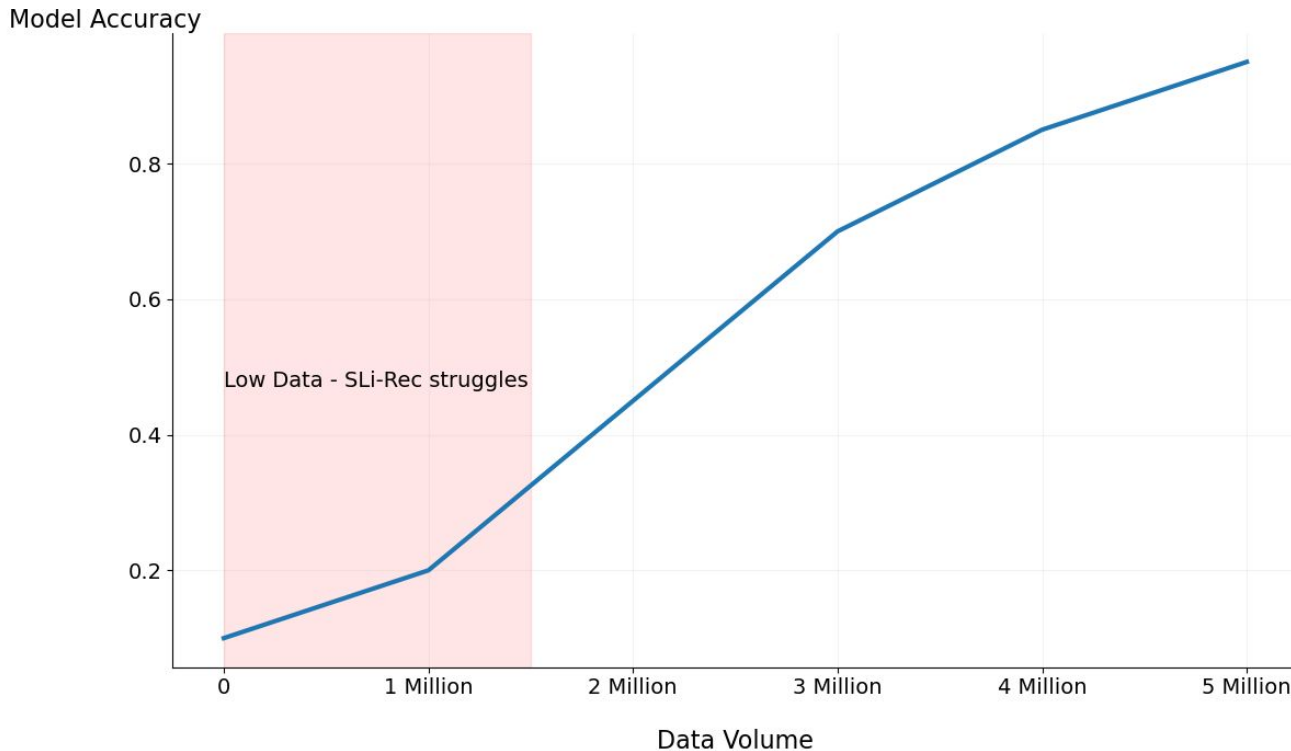


SLi-Rec performs well with large data volume



SLi-Rec being a neural network based recommendation system, excels with large data volumes.

SLi-Rec struggles with low data volume



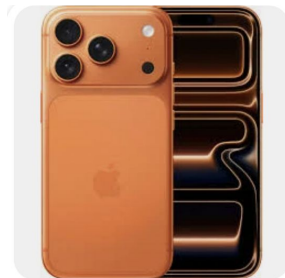
However, it has an issue with low data availability. The required high data volumes may not always be available.

In this scenario, we can **increase randomness** to facilitate **novel recommendations** until we have more data available on the product

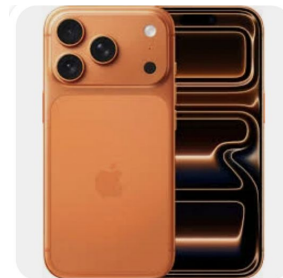
Sequential Recommendations



Apple - iPhone 17 Pro 1TB - Cosmic Orange (Verizon) :
\$1,499.99



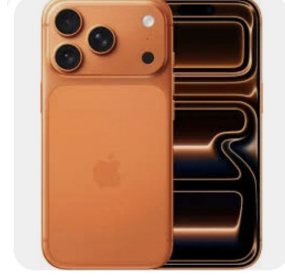
Apple - iPhone 17 Pro 1TB - Cosmic Orange (Verizon) :
\$1,499.99



Apple - iPhone 17 Pro 1TB - Cosmic Orange (Verizon) :
\$1,499.99



Samsung Galaxy Z Flip7 FE :
\$1,093.14



Apple - iPhone 17 Pro 1TB - Cosmic Orange (Verizon) :
\$1,499.99



Google Pixel 10 Pro Factory Unlocked :
128gb - Obsidian, Black
\$939.97

Sequential Recommendations

THANK YOU!